

A
SEMINAR REPORT ON

“Humanoid Robot”

Submitted in partial fulfilment of requirements for the award of

degree of

Bachelor of Technology

in

Electronics and Telecommunication Engineering

by

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under the guidance of

Prof. Md. Sadique Sheikh



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Maharashtra State, India 2022-2023

CERTIFICATE

This is to certify that the Seminar report entitled,

“Humanoid Robot ”

is a bona fide Seminar work and has been carried out by team:

Shubham Naresh rathod:- (2210121372034)

of first year B-Tech class under the guidance of **Prof. Md. Sadique Sheikh** during the academic year 2022 -2023 (Sem-II).

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Seminar Approval Sheet

This is to certify that the Seminar report entitled,

“Humanoid Robot”

is a bona fide Seminar work submitted by:

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Electronics and Telecommunication Engineering.

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DECLARATION

I declare that this Seminar report titled **Humanoid Robot** submitted in partial fulfilment of the degree of B. Tech in Electronics and Telecommunication Engineering is a record of original work carried out by me under the supervision of **Prof. Md. Sadique Sheikh** and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgement have been made wherever the findings of others have been cited

I, Shubham Rathod, hereby declare that the following document represents my original work and is the result of my own research efforts. This Seminar submission titled "**Humanoid Robot**" has been prepared in partial fulfilment of the requirements for the completion of B. Tech Electronics and Telecommunication.

I affirm that all the information presented in this Seminar is based on authentic sources and has been appropriately referenced and cited. Any external material or sources used have been duly acknowledged and credited. I have taken all necessary measures to ensure the accuracy and reliability of the data and information included in this Seminar.

I further declare that this Seminar represents my independent work and that no part of it has been submitted for any other purpose or assessment. The ideas, concepts, methodologies, and findings presented in this document are solely my own, unless otherwise stated and referenced.

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ACKNOWLEDGEMENT

A part from individual efforts, the success of any Seminar depends largely on the encouragement and guidelines of many others. We take this opportunity to express our gratitude to the people who have been instrumental throughout the Seminar work. It is our privilege to express our gratitude towards our Seminar guide, **Prof. Md Sadique Sheikh**, for his valuable guidance, encouragement, inspiration and wholehearted cooperation throughout the Seminar work. We thank him for being a motivation through all our highs and importantly, our lows.

We deeply express our sincere thanks to our Head of department Prof. Md Sadique Sheikh for encouraging and allowing us to present the Seminar on the topic “Humanoid Robot ” and providing us with the necessary facilities to enable us to fulfil our Seminar requirements as best as possible. We take this opportunity to thank all faculty members and staff of the Electronics and Telecommunication Engineering department who have directly or indirectly helped us in our Seminar.

We pay our respects to honourable Principal **Dr. P. M. Khodke** for his encouragement and motivation. Our thanks and appreciations also go to our family and friends, who have been a source of encouragement and inspiration throughout the duration of the Seminar.

ABSTRACT

In rapid change with technology a huge development of robots is competing equally with humans. The technology development has brought many changes in the field of manufacturing and research. Initially, the robots were used by humans to make their work easier, but today the robots are developed to replicate humans and substitute the labor in the fields where even human presence is difficult. The development doesn't only take place in the manufacturing of robots but simultaneously it focuses on control and optimization of the tasks that are done by the robots. The aspects of robots are unlimited in the field of medical, defense, education. The type of application defines the robots construction and the requirement of systems. This paper covers background of the robotics such as variety of robots, kinematics of robots, localization and mapping scenario, the trajectory of the robot's path. Also, the control systems used in robots like actuators, sensors and processing of the actuators, robots learning has been presented. The experiment with humanoid robots for civilians has designed through programming and controlled by microcontrollers for analyzing the results of degrees of freedom and positions of robots.

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CHAPTER 1

INTRODUCTION

A humanoid robot is an advanced machine designed to resemble and interact with humans in both appearance and behavior. It is a technological marvel that combines the fields of robotics, artificial intelligence (AI), and engineering to create a lifelike entity capable of performing various tasks. These robots are built with a human-like body structure, including a head, torso, arms, and legs, enabling them to move and function in a manner similar to humans.

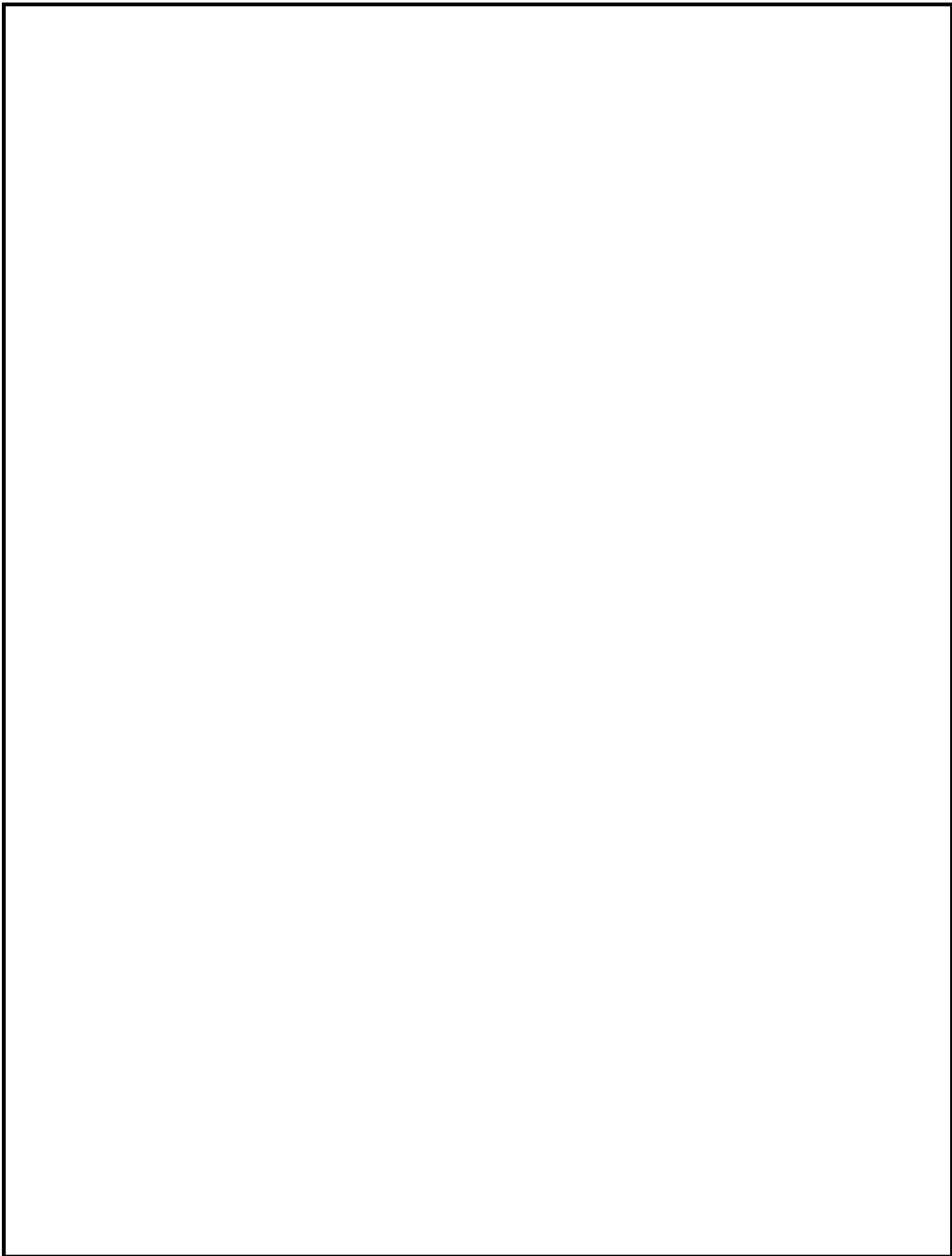
The development of humanoid robots has been driven by the desire to create machines that can assist humans in a wide range of activities, such as household chores, healthcare, manufacturing, and entertainment. By imitating human form and movement, these robots can navigate human environments more effectively and interact with people in a more intuitive manner.

One of the key components of humanoid robots is their AI systems. They are equipped with sophisticated algorithms and sensors that allow them to perceive and understand the surrounding environment. Through advanced image processing and voice recognition techniques, these robots can interpret visual and auditory stimuli, enabling them to recognize faces, understand speech, and respond appropriately.

The locomotion capabilities of humanoid robots are achieved through intricate mechanical systems and advanced control algorithms. They are equipped with joints and actuators that replicate human movement, allowing them to walk, run, pick up objects, and perform various other physical tasks. Additionally, some humanoid robots have the ability to mimic human facial expressions, gestures, and body language, enhancing their ability to communicate with humans effectively.

The potential applications of humanoid robots are vast and diverse. In healthcare, they can assist with patient care, rehabilitation exercises, and surgery. In manufacturing, they can perform complex tasks on assembly lines, increasing efficiency and productivity. In the entertainment industry, they can entertain and engage audiences with their lifelike performances. Moreover, humanoid robots have the potential to act as companions for the elderly or individuals with disabilities, providing social interaction and assistance in daily activities.

However, despite significant advancements, humanoid robots still face challenges. Achieving seamless human-like movement, ensuring robustness and reliability, and addressing ethical and safety concerns are ongoing areas of research and development. Nonetheless, with continued progress, humanoid robots hold immense promise in revolutionizing various aspects of our lives and shaping the future of robotics and AI.



CHAPTER 2

Methodology

2.1 Sensors

A [sensor](#) is a device that measures some attribute of the world. Being one of the three primitives of robotics (besides planning and control), sensing plays an important role in [robotic paradigms](#). Sensors can be classified according to the physical process with which they work or according to the type of measurement information that they give as output. In this case, the second approach was used.

2.2 Proprioceptive:

[Proprioceptive](#) sensors sense the position, orientation, and speed of the humanoid's body and joints, along with other internal values.

In human beings, the otoliths and semi-circular canals (in the inner ear) are used to maintain balance and orientation. Additionally, humans use their own proprioceptive sensors (e.g. touch, muscle extension, limb position) to help with their orientation. Humanoid robots use [accelerometers](#) to measure the acceleration, from which velocity can be calculated by integration; [tilt sensors](#) to measure inclination; force sensors placed in robot's hands and feet to measure contact force with environment; position sensors that indicate the actual position of the robot (from which the velocity can be calculated by derivation); and even speed sensors.



Figure 2.2.1 An artificial hand holding a lightbulb

Arrays of [tactels](#) can be used to provide data on what has been touched. The [Shadow Hand](#) uses an array of 34 tactels arranged beneath its [polyurethane](#) skin on each finger tip.[22] Tactile sensors also provide information about forces and torques transferred between the robot and other objects.

Vision refers to processing data from any modality which uses the electromagnetic spectrum to produce an image. In humanoid robots it is used to **recognize objects** and determine their properties. Vision sensors work most similarly to the eyes of human beings. Most humanoid robots use **CCD** cameras as vision sensors. Sound sensors allow humanoid robots to hear speech and environmental sounds, akin to the ears of the human being. **Microphones** are usually used for the robots to convey speech.

2.3 Actuators

Actuators are the motors responsible for motion in the robot.

Humanoid robots are constructed in such a way that they mimic the human body. They use actuators that perform like **muscles** and **joints**, though with a different structure. The actuators of humanoid robots can be either electric, **pneumatic**, or **hydraulic**. It is ideal for these actuators to have high power, low mass, and small dimensions.

2.4 Electric

Electric actuators are the most popular types of actuators in humanoid robots. These actuators are smaller in size, and a single electric actuator may not produce enough power for a humansized joint. Therefore, it is common to use multiple electric actuators for a single joint in a humanoid robot. An example of a humanoid robot using electric actuators is **HRP-2**.

2.5 Hydraulic

Hydraulic actuators produce higher power than electric actuators and pneumatic actuators, and they have the ability to control the torque they produce better than other types of actuators. However, they can become very bulky in size. One solution to counter the size issue is **electro-hydrostatic actuators** (EHA). The most popular example of a humanoid robot using hydraulic actuators is the **ATLAS** robot made by **Boston Dynamics**.

2.6 Pneumatic

Pneumatic actuators operate on the basis of **gas compressibility**. As they are inflated, they expand along the axis, and as they deflate, they contract. If one end is fixed, the other will move in a linear **trajectory**. A popular example of a pneumatic actuator is the **Mac Kibben muscle**

2.7 Planning and control

Planning in robots is the process of planning out motions and trajectories for the robot to carry out. Control is the actual execution of these planned motions and trajectories. In humanoid robots, the planning must carry out biped motions, meaning that robots should plan motions similar to a human. Since one of the main uses of humanoid robots is to interact with humans, it is important for the planning and control mechanisms of humanoid robots to work in a variety of terrain and environments.

The question of walking biped robots stabilization on the surface is of great importance. Maintenance of the robot's gravity center over the center of bearing area for providing a stable position can be chosen as a goal of control.

To maintain dynamic balance during the **walk**, a robot needs information about contact force and its current and desired motion. The solution to this problem relies on a major concept, the **Zero Moment Point** (ZMP). Another characteristic of humanoid robots is that they move, gather information (using sensors) on the "real world", and interact with it. They do not stay still like factory manipulators and other robots that work in

highly structured environments. To allow humanoids to move in complex environments, planning and control must focus on self-collision detection, [path planning](#) and [obstacle avoidance](#).^{[29][30]}

Humanoid robots do not yet have some features of the human body. They include structures with variable flexibility, which provide safety (to the robot itself and to the people), and redundancy of movements, i.e. more [degrees of freedom](#) and therefore wide task availability. Although these characteristics are desirable to humanoid robots, they will bring more complexity and new problems to planning and control. The field of whole-body control deals with these issues and addresses the proper coordination of numerous degrees of freedom, e.g. to realize several control tasks simultaneously while following a given order of priority.

CHAPTER 3 ADVANTAGES AND DISADVANTAGES

3.1 Advantages of Humanoid Robot

It has a head designed to replicate human facial features like the eyes and mouths, It can be used for functional purposes, like interacting with human tools & environments, for experimental purposes, such as the study of locomotion, or for other purposes.

[Humanoid robots](#) can perform any task a human being can, so long as they have the proper software, Although they look realistic, they have no cognition or physical autonomy, They can be used for future dangerous & distant space exploration missions, They can serve as companions & assistants for humans in daily life, they can be used as ultimate helpers in man-made & natural disasters.



3.1.1 Humanoid robots

[Humanoid robots](#) will be useful to many people around the globe, We can use these robots for educational purposes as well, [Robotics](#) is being used by teachers to help instruct students on how to program, [Robot teacher](#) is also used to interact with children and teach them to do simple tasks.

Many people can benefit from these [humanoid robots](#), They will take care of the sick, collect the rubbish, guard homes & offices, and give directions on the street, Life with [Humanoid Robots](#) in society will be very different but very efficient and the economy will be increased. Corporations and businesses will also be appreciative of [the robots](#) because they could use them to replace jobs, This means less money that businesses will have to fork out to their employees, They can be used as entertainers & in commercial spaces but they will not undertake major tasks, These [robots](#) can do amazing things that even people could not do, They can now sing & dance and look almost like the human when doing so.

Most artificial intelligence does not need a body to work, the human body is very hard to replicate and there is much more work to be done, They can be useful in helping elderly people live in their own homes to respond to disasters, There are great developments & rapid advancement in the engineering and control aspects of the [humanoid robotics](#). [Humanoid robots](#) can be used to help the military, They can move, gather information (using sensors) on the real world and interact with it, They do not yet have some features of the human body, They include structures with variable flexibility, which offer safety (to the robot itself and to the people), & redundancy of movements and Law enforcement will benefit from the [humanoid robots](#).

Humanoid Robots can take care of the elderly and they can be programmed to ensure that they will not be disrespectful, as well as be around an elderly person full time and be able to care for their needs and safety, They do not make mistakes is also beneficial in the workplace.

Humanoid Robots can work in factories and do the same thing over & over again and without any difference, Humans, on the other hand, will get tired after doing the same thing repetitively and could easily make a mistake.

Humanoid Robots are able to do many things that people can do and even things that people are unable to do, The sale & development of these **Humanoid Robots** will help increase the economy, Having these robots can make companies more efficient in their work and spur the economy with their revenue growth.

Humanoid Robots do not make the mistakes that are useful to some countries, They prefer to have the robots do menial jobs rather than low waged immigrants from other countries, The people prefer **the robots** because they do not make mistakes and there is no chance for them to be disrespectful.

Humanoid Robots can be used with children, They can be used to teach children or read to them, They can help children with autism immensely, They can assist the sick & elderly, They can be used in dirty or dangerous jobs, They are suitable for some procedurally-based vocations, such as reception-desk administrators and **automotive manufacturing** line workers, they can use the tools and operate equipment and vehicles designed for the human form.

3.2 Disadvantages of Humanoid Robots

A **humanoid robot** is very expensive, depending on how advanced the robot is, As **humanoid robot** starts becoming more and more available to society, it will be mostly the wealthy who will be able to afford them, It will replace the jobs, Although this will be beneficial to many companies, it will hurt those who have jobs in certain fields, such as nurses, pharmacists, secretaries, etc.

The advanced **humanoid robots** will be able to complete a variety of tasks, but they cost a lot of money, For middle-class & low-income families, It is a high expense that many people can't afford, The price of humanoid robots is high enough that everyone can't own one, one of the common problems tackled in the **humanoid robotics** is the understanding of the human-like information processing and the underlying mechanisms of the human brain in dealing with the real world.

If you have these robots doing these everyday jobs for you, You will become too dependent on robots, If you have robots that will take out the trash, do the dishes, cook the food, and watch the children, You will become lazy, and dependent if you have such robots doing your chores for you, This is another downfall of the evolving "humanoid robot."

A lot of people will suffer from the **humanoid robot**, Humanoid robot will replace their jobs, and many jobs can be replaced by automation systems & robots, The humanoid robots will one day replace the jobs of middle-class people, They will take nearly all traditional labor roles in the society like **firefighting**, **food service**, **manufacturing**, **agriculture**, construction, & community policing, If the humanoid robots take over middle-class citizens jobs, our economy will be negatively affected.

CHAPTER 4

The Future Scope of Humanoid Robots



FIGURE 4.1.1

The scope of humanoid robots goes beyond just being auxiliary machines or companion robots

While humanoids currently are one of the smallest groups of service robots, they have the potential to become the industrial tool of the future. The robotics of tomorrow is an exciting prospect as companies like Softbank Robotics and others have created human-like robots to be used as medical assistants and teaching aids.

Currently, humanoid robots are outshining in the medical industry, particularly as companion robots. Even, companies are leveraging these kinds of robots to fill engineering tasks. For instance, a four-year joint research project was conducted by Joint Robotics Laboratory and Airbus Group to make use of humanoid robotic technology in aircraft manufacturing facilities. By utilizing humanoid robots on aircraft assembly lines, Airbus looks to free up human operators of some of the most arduous and perilous tasks, so that they could then focus on higher-value tasks.

However, the prime intricacy found with these robots is the confined spaces where they have to work in as well as being able to move without colliding with the surrounding objects.

While the robotic industry will witness an increase in humanoid robots, it will also affect the service industry. Looking at industry insights, the robots in the service industry are largely led by floor cleaning robots; unmanned aerial vehicles (UAVs) and lawnmowers; and humanoid robots, assistant robots and companion robots, telepresence robots, human exoskeletons, surgical robots and autonomous mobile robots.

Many countries like the US, UK, Canada, Japan, among others will be leading the adoption of humanoid robots. Also, the four major industries that will be seen taking charge of the advancements would be computer and electronic products, the electrical equipment and appliances industry, transportation equipment, and machinery.

Thus, there are much larger expectations that humanoid robots in years to come will serve as companions and assistants to humans in their daily lives. And ultimately, they will give their helping hands in times of natural disasters.

CHAPTER 5 Conclusion

The application of humanoid robots is crucial in the field of healthcare, education and as a social robot. In all these domains, most of the research is focused on the effect of using the humanoid robot, user acceptance and trust on the robot. It can be observed from the existing literature that opinion of the people towards the use of humanoid robot varies from individual to individual. A user of different age, gender, and health condition perceive the usage and importance of humanoid robots differently. The appearance of the robot was found to be a crucial factor affecting user acceptance and trust; moreover, robot functionality and gaze also affected user preference towards humanoid robot application.

Although many studies have been conducted to analyze the usage and effectiveness of humanoid robots, we still need more research in this field. As future research, the influence of incorporating a humanoid robot into the Project Leonard as an assistant to care managers in order to enhance disease and care management effect must be addressed

Reference

Humanoid Robotics: a Reference provides a comprehensive compilation of developments in the conceptualization, design and development of humanoid robots and related technologies. Human beings have built the environment they occupy (living spaces, instruments and vehicles) to suit two-legged systems. Building systems, especially in robotics, that are compatible with the well-established, human-based surroundings and which could naturally interact with humans is an ultimate goal for all researchers and engineers. Humanoid Robots are systems (i.e. robots) which mimic human behavior. Humanoids provide a platform to study the construction of systems that behave and interact like humans. A broad range of applications ranging from daily housework to complex medical surgery, deep ocean exploration, and other potentially dangerous tasks are possible using humanoids. In addition, the study of humanoid robotics provides a platform to understand the mechanisms and offers a physical visual of how humans interact, think, and react with the surroundings and how such behaviors could be reassembled and reconstructed. Currently, the most challenging issue with bipedal humanoids is to make them balance on two legs, The purportedly simple act of finding the best balance that enables easy walking, jumping and running requires some of the most sophisticated development of robotic systems- those that will ultimately mimic fully the diversity and dexterity of human beings. Other typical human-like interactions such as complex thought and conversations on the other hand, also pose barriers for the development of humanoids because we are yet to understand fully the way in which we humans interact with our environment and consequently to replicate this in humanoids.